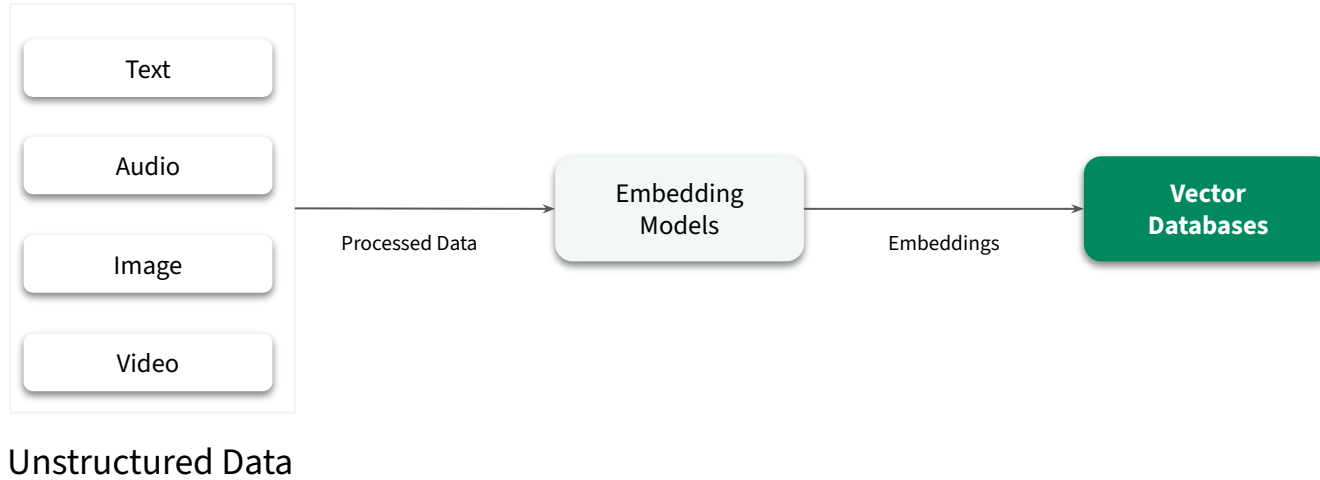


Introduction

Vector Databases



Vector Databases: **The database for LLMs**



The Search for Similar

Vector Databases



Embeddings: **Semantic Meaning** –> **Concept**

Traditional Search

Keyword: Dog

Lexical Matches

Dog

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From Wikipedia, the free encyclopedia

"Doggy" and "Pooch" redirect here. For other uses, see [Dog \(disambiguation\)](#), [Doggy \(disambiguation\)](#), and [Pooch \(disambiguation\)](#).

The **dog** (***Canis familiaris*** or ***Canis lupus familiaris***) is a [domesticated](#) descendant of the [gray wolf](#). Also called the **domestic dog**, it was [selectively bred](#) from a population of wolves during the [Late Pleistocene](#) by [hunter-gatherers](#). The dog was the first [species](#) to be domesticated by [humans](#), over 14,000 years ago and before the [development of agriculture](#). Due to their long association with humans, dogs have gained the ability to thrive on a [starch-rich](#) diet that would be inadequate for other [canids](#).

Dogs have been bred for desired behaviors, sensory capabilities, and physical attributes. [Dog breeds](#) vary widely in shape, size, and color. They have the same number of bones (with the exception of the tail), powerful jaws that house around 42 teeth, and well-developed [senses of smell](#), hearing, and sight. Compared to humans, dogs possess a superior sense of smell and hearing, but inferior [visual acuity](#). Dogs perform many roles for humans, such as [hunting](#), [herding](#), [pulling loads](#), [protection](#), [companionship](#), [therapy](#), [aiding disabled people](#), and [assisting police](#) and the [military](#).

[Communication in dogs](#) includes eye gaze, facial expression, vocalization, body posture (including movements of bodies and limbs), and gustatory communication (scents, pheromones, and taste). They mark their territories by urinating on them, which is more likely when entering a new environment. Over the millennia, dogs have uniquely adapted to human behavior: this adaptation includes being able to understand and communicate with

Dog

Temporal range: 0.0142–0 Ma

PreC

P

S

D

C

T

J

K

Pch

Late Pleistocene (14,200 years ago) to present^[1]

Conservation status

Domesticated

Scientific classification

Kingdom: [Animalia](#)



Keyword: Dog

[Dog](#)
[🔍](#) [📄](#) [🗨️](#) [👤](#) [🌐](#)

[Dog](#)
[Talk](#)

[Read](#)
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The dog (***Canis familiaris*** / ***canis lupus familiaris***) is a domesticated descendant of the gray wolf. Also called the **domestic dog**, it was selectively bred from a population of wolves during the Late Pleistocene by hunter-gatherers. The dog was the first species to be domesticated by humans, over 14,000 years ago and before the development of agriculture. Due to their association with humans, dogs have gained the ability to live on a starch-rich diet that would be inadequate for other canids.

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Communication in dogs includes eye gaze, facial expression, vocalization, body posture (including movements of bodies and limbs), and gustatory communication (scents, pheromones, and taste). They mark their territories by urinating on them, which is more likely when entering a new environment. Over the millennia, dogs were uniquely adapted to human behavior; this adaptation includes being able to understand and communicate with

Semantic Matches

[illegible]

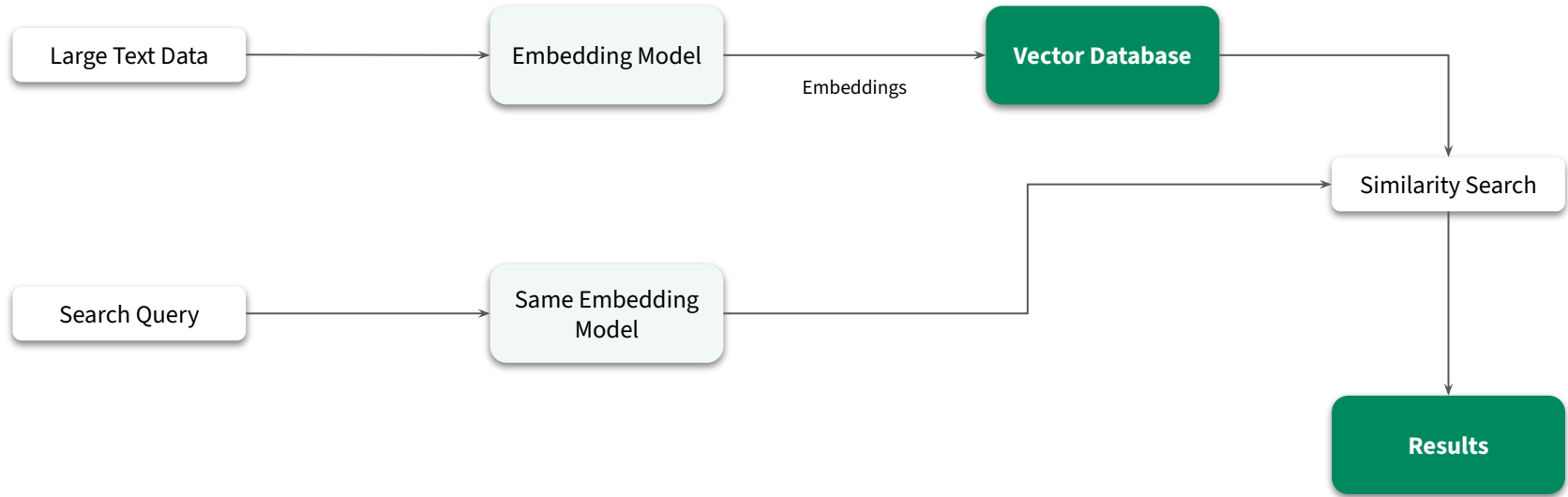


Traditional Search: **Keywords**

Semantic Search: **Concepts**

FAISS

Vector Databases

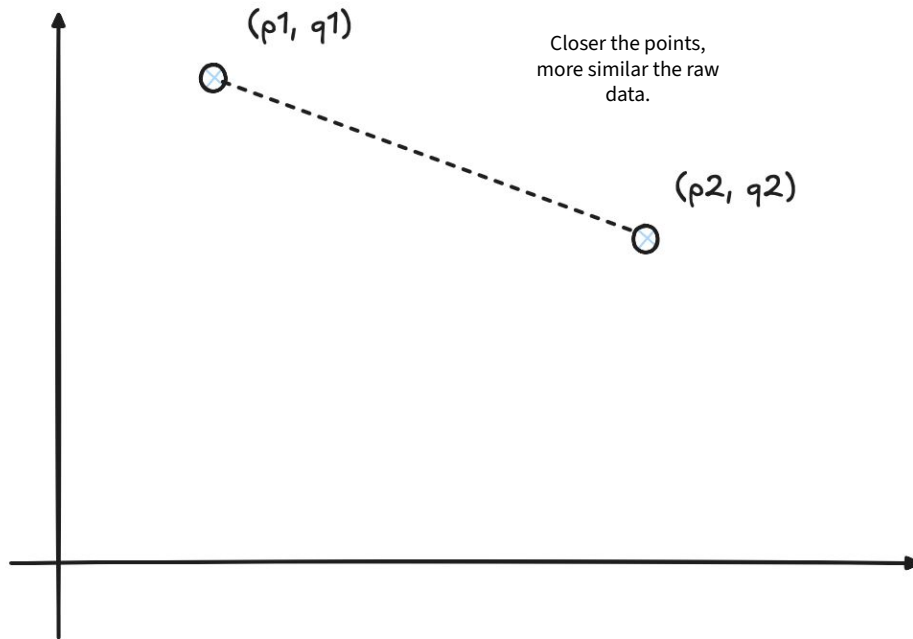


Search Metrics

Vector Databases

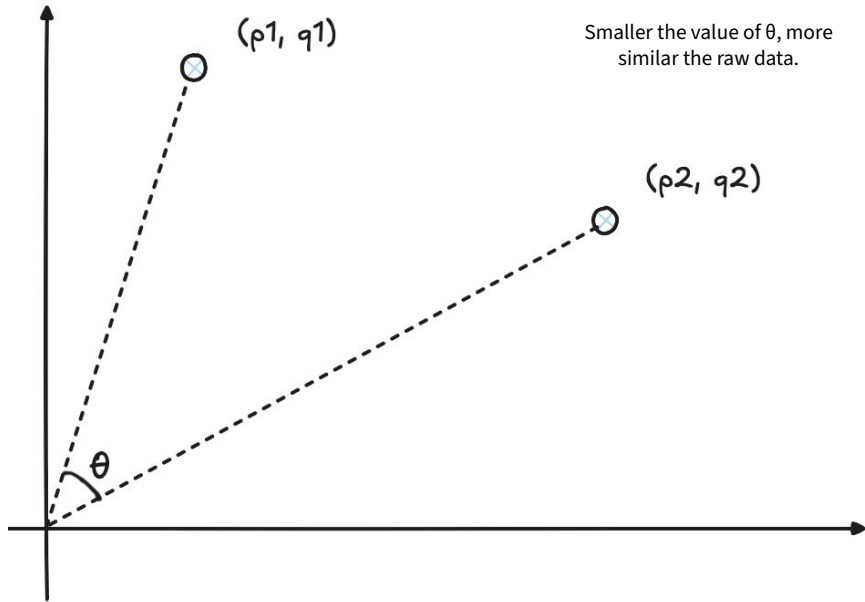
Euclidean Distance

Shortest straight line between two vectors.



Cosine Similarity

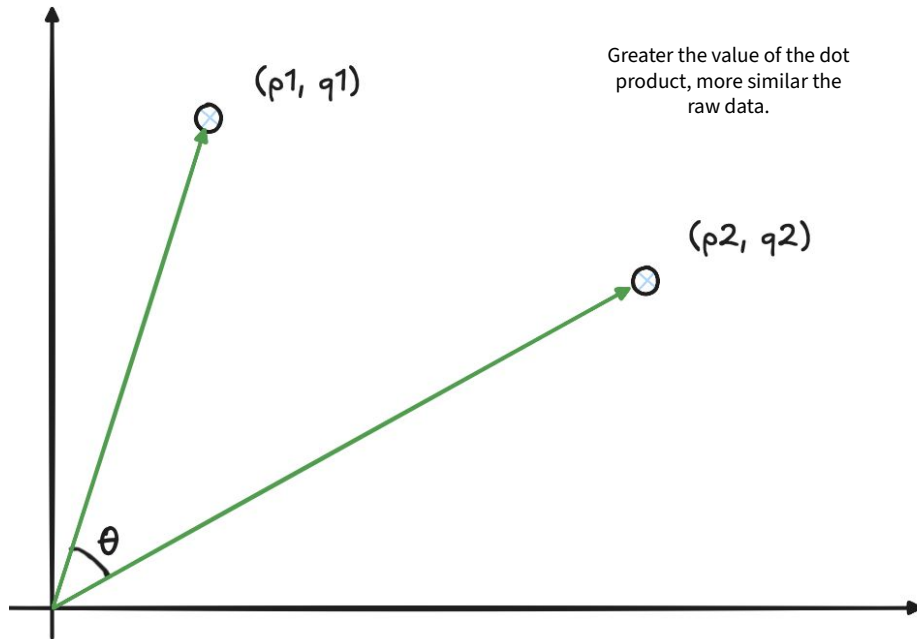
Angle between two vectors.





Dot Product

Angle and Magnitude between two vectors.





When to use which similarity metric?

Metric Type	Common Use Cases
Euclidean Distance	Image Recognition, Similarity & Search
Cosine Similarity	Text Similarity & Search
Dot Product	Recommendation Systems

Search Mechanisms

Vector Databases



Exact Neighbour Search

Exhaustive, accurate search.



When to use which search mechanism?

3S: **S**peed, **S**cale and **S**pecificity (accuracy)

Search Mechanism	Pros	Cons	When to Use	Use Cases
Exact Neighbour Search	Perfectly accurate; Simple to understand.	Extremely Slow	Choose when accuracy is paramount, and application does not need real-time results	Scientific research, medical diagnostics, small-scale academic projects.



Approximate Nearest Neighbor Search

Fast, group-based search.



When to use which search mechanism?

3S: **S**peed, **S**cale and **S**pecificity (accuracy)

Search Mechanism	Pros	Cons	When to Use	Use Cases
Exact Neighbour Search	Perfectly accurate; Simple to understand.	Extremely slow	Choose when accuracy is paramount, and application does not need real-time results	Scientific research, medical diagnostics, small-scale academic projects.
Approximate Neighbour Search	Extremely fast and scalable	Slight accuracy loss	Choose when closest relevant results work, <u>or</u> application may have real-time requirements	Image search, recommendation engines, semantic search, anomaly detection



Hybrid Search

Combine keyword & vector search results.



When to use which search mechanism?

3S: **S**peed, **S**cale and **S**pecificity (accuracy)

Search Mechanism	Pros	Cons	When to Use	Use Cases
Exact Neighbour Search	Perfectly accurate; Simple to understand.	Extremely slow	Choose when accuracy is paramount, and application does not need real-time results	Scientific research, medical diagnostics, small-scale academic projects.
Approximate Neighbour Search	Extremely fast and scalable	Slight accuracy loss	Choose when closest relevant results work, <u>or</u> application may have real-time requirements	Image search, recommendation engines, semantic search, anomaly detection
Hybrid Search	Enhanced accuracy and relevance; Handles a wider range of queries	Increased complexity; Higher computational and resource costs	Choose when user intent is complex and can benefit from both keyword precision and semantic understanding.	Modern search engines, QnA chatbots

Popular Vector DBs

Vector Databases



Milvus DB

Open Source

Introduced in 2019

High Performance

Use GPU and other hardware to support trillion-vector datasets

Advanced Indexing

Offers multiple powerful indexing approaches

Scalability & Elasticity

Separates storage and compute allowing for segregated scaling

Unified API

Intuitive code for multiple languages along with a RESTful API.

Hybrid Search

Native hybrid search features available



PGVector

Open Source Extension

Introduced in 2021 for PostgreSQL

Simplicity and Convenience

No need to learn another database; just use PostgreSQL

Cost-Effective

Eliminates needs of multiple database setups

Limited Scalability

Not designed for massive, distributed scale

Fewer Advanced Features

Lacks hardware optimisation, index fine-tuning, etc

ACID Compliant

Transaction datasets fit naturally



PineCone

Fully Managed

Setup, maintenance, scaling, or infrastructure management handled

High Performance and Low Latency

Optimized for real-time search with low query latency, even at scale

Ease of Use

Extremely simple; Managed model reduces operational overhead

Hybrid Search

Native hybrid search features available

Scalability

Designed to scale seamlessly without performance degradation

Serverless Experience

Users only need to worry about their data and queries, not the underlying servers.



Chroma

Open Source

Released in 2022

Developer-First API

Intuitive Python API that is easy to learn and integrate

Native Embeddings

Includes built-in support for embedding functions from OpenAI, Cohere, and Hugging Face

Flexible Deployment

Can be run in-memory (e.g., in a Jupyter notebook), as a persistent database on a single machine, or in a client/server mode

Analysis and Visualization

Provides utilities to help developers understand and analyze their embeddings

Great Integrations

Deeply integrated with popular LLM framework like LangChain



Qdrant

Open Source

Written in Rust for maximum speed and efficiency

Advanced Filtering

Complex queries can be written easily

Payload Storage

Stores not just the vectors but also associated JSON payloads (metadata), which can be filtered upon

On-Disk Storage

Designed with on-disk storage in mind, allowing it to handle datasets larger than the available RAM

Quantization

Provides utilities to help developers understand and analyze their embeddings

Flexible Deployment

Available as open-source, a Docker image, and a managed cloud solution